

Josh Loecker

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PROFESSIONAL SUMMARY

I am a computational biologist with **expertise in software development, bioinformatics, metabolic modeling**, and RNA-seq analysis. Most of my experience is in building maintainable backend software, deploying scalable high-performance computing solutions, and leading cross-disciplinary teams. I am very **passionate about developing robust and maintainable software that is usable by individuals without computational expertise**.

TECHNICAL SKILLS

Software Design & Engineering: Python, R, Snakemake, SQL, Object Oriented Programming

Bioinformatics: Bulk & single-cell RNA-seq, metabolic modeling, multi-omics integration

Systems & Tools: Docker, Git, Linux administration, CI/CD, cloud & cluster computing

AI/ML: Large Language Models (LLMs), integration with computational modeling and biological data

EXPERIENCE

Graduate Student and Postdoctoral Research Associate (Biochemistry, Bioinformatics)

PI: Dr. Tomáš Helikar; University of Nebraska-Lincoln

August 2021 to Present

- **MechAInistic:** AI agents design and execute in-silico experiments on constraint-based metabolic models
 - **Integrates 30 custom-made tools** for LLM-based function calling
 - Retrieves information from PubMed Central for up-to-date biomedical information
 - Connects to any OpenAI-compatible backend
 - **Multi-LLM architecture** attempts to reduce hallucinations
 - *Preprint: <https://doi.org/10.64898/2026.05.11.723319>*
- **COMO:** Constraint-based Optimization of Metabolic Objectives
 - Automated **reconstruction of constraint-based metabolic models**
 - Incorporates **multi-omics integration** of proteomics, bulk, and single-cell RNA-sequencing data
 - Reduced Docker **build time from 4 hours to 20 minutes**
 - Provides a Python-based API and Jupyter Notebook frontend
 - *Manuscript: <https://doi.org/10.1093/bib/bbad387>*
- **AutoRNAseq:** Automated alignment of bulk RNA-sequencing samples
 - Automatically **downloads any reference genome** from Ensembl
 - **Built-in quality control** analysis on a per-sample basis
 - Dynamic deployment options allow for a local desktop or remote compute cluster execution
 - *Preprint: <https://doi.org/10.64898/2026.04.21.719844>*
- **Single-cell Analysis**
 - **Reconstruct 5,705 constraint-based metabolic models** with patient-specific scRNA transcriptomics
 - Created a generic **metabolic model validation framework**
 - Identified metabolic pathways significantly correlating with age in specific sub-cell-types
- **Other**
 - **Supervised a team of undergraduates** by assigning tasks, reviewing code, mentoring development, and deploying production systems
 - **Managed high-performance computing workflows** on a Linux-based supercomputing cluster.

Bioinformatics Technician

PI: Dr. Patrick Ewing; U.S. Department of Agriculture

May 2020 to May 2021

- **Built an advanced RNA-sequencing pipeline** using high-performance computing to reduce manual labor required in downstream data analysis
- **Built and deployed specialized soil sieves** for bacterial analysis of pre- and post-crop season using a variety of fertilizers, herbicides, and pesticides.

Lab Technician

PI: Dr. Surtaj Iram; South Dakota State University

August 2018 to May 2020

- **Created a graphical user interface** in Python to automate data analysis
- Performed **MTT cell-viability assays** to determine the drug efficacy of multidrug-resistant cell lines
- Performed **ELISA assay techniques** for accurate antibody detection

EDUCATION

Doctor of Philosophy in Biochemistry with Specialization in Bioinformatics

University of Nebraska-Lincoln, July 2026

Bachelor of Science in Biochemistry, Minor in Computer Science

South Dakota State University, Cum laude, May 2021

AWARDS & PRESENTATIONS

- Oral Presentation, Department of Biochemistry Department Retreat, Lincoln, NE August 2025
- Oral Presentation, Society for Mathematical Biology, Edmonton, Canada July 2025
- Poster Presentation, Systems Immunology, Cold Spring Harbor, NY April 2025
- Oral Presentation, Molecular Mechanisms of Disease, Lincoln, NE April 2023 & April 2024
- Poster Presentation, "Building an Immune Digital Twin", Orsay, France May 2023
- Molecular Mechanisms of Disease Award Recipient January 2022–January 2024
- South Dakota State University Chemistry Club President August 2019–May 2021
- East Family Undergraduate Summer Research Fellowship Summer 2019 & Summer 2020
- BioSNTR Undergraduate Summer Research Fellowship Summer 2018
- Herbert Hodgeson Scholarship in Chemistry Spring 2018

PUBLICATIONS (*Authors contributed equally to this work)

- **Loecker J**, Pujara N*, Bryant W*, Puniya BL, Packrisamy P, Hamed AA, Helikar T. MechAInistic: An LLM-guided Multi-Agent System for Reasoning over Genome-Scale Constraint-Based Metabolic Models. bioRxiv. 2026. 10.64898/2026.05.11.723319.
- **Loecker J**, Bessell B, Puniya B, Helikar T. AutoRNAseq: Automated Bulk RNA-seq Analysis Pipeline. bioRxiv. 2026. 10.64898/2026.04.21.719844.
- Bessell B*, **Loecker J***, Zhao Z, Aghamiri S, Mohanty S, Amin R, et al. COMO: A pipeline for multi-omics data integration in metabolic modeling and drug discovery. Brief Bioinform. 2023. 10.1093/bib/bbad387.
- **Loecker J**, Ewing P. Benefits of the Snakemake Workflow Management Software in Comparison to Traditional Programming. *Undergraduate Honors Independent Study*. 2021. https://openprairie.sdstate.edu/honors_isp/8.